

Governance Quality Tests 41–45

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

This note presents five binary governance quality tests — Tests 41 through 45 — for Full Aspect Integration (FAI) governance review. Together these tests address governance quality properties beyond structural compliance: determinism contract satisfaction (Test 41), documentation standards compliance (Test 42), four-link authorization chain completeness (Test 43), non-specialist accessibility (Test 44), and non-delegation compliance (Test 45). Each test specifies the records to examine, the binary question that decides the outcome, what a PASS indicates, and what a FAIL indicates along with the anti-pattern triggered. The note explains why quality tests occupy a distinct audit layer from structural compliance tests: a governance record can exist and still fail every quality test, and both layers are necessary for a comprehensive audit. It also explains why Test 41 functions as a governance quality meta-test, and why Test 44 connects governance quality to the operational validity of the accessibility commitment the architecture makes at design time.

1. Why governance quality tests are a distinct audit layer

Structural compliance tests ask whether governance structures exist. They verify that FAI configuration records are present, that conflict routing rules have been produced, that resolution records accompany resolved conflicts, and that the other structural artifacts the architecture requires have been created. A structural compliance test that passes establishes that the governance forms are in place.

Governance quality tests ask something different: whether the governance structures that exist meet quality standards. The distinction matters because a governance record can satisfy every structural requirement while failing substantive quality requirements. Configuration records can exist but be incomplete in ways that make decisions unreproducible. Documentation can satisfy format requirements but fail provenance standards. Authorization chains can appear to run from outcome to originating

decision but break upon inspection at one of the intermediate links. Governance records can be present and formally correct while expressed in technical vocabulary that non-specialist practitioners cannot evaluate. Decision records can exist while covering only automated executions without the human pre-authorizations those automations were required to reference.

None of these failures is visible to a structural compliance test. They are visible only to tests that examine quality properties of existing records. The two audit layers are therefore complementary and both necessary: structural compliance establishes that the governance infrastructure exists; quality tests establish that the existing infrastructure is substantively fit for the governance purposes it is supposed to serve.

The five tests in this note — Tests 41 through 45 — constitute Expansion Category 3 within the D5 operational test series for Paper 3 commitments. They are specific to FAI governance and draw on commitments formalized across the derivation note series: the determinism contract (D2.66), documentation standards (D2.36), the four-link authorization chain (D2.67), non-specialist governance (D2.75), and non-delegation (D2.43).

2. Test 41 — Determinism Contract Satisfaction

Records to examine: FAI configuration records; conflict routing rules; resolution records; evolution feed configuration records; amendment version history.

Binary question: Can all five determinism requirements be verified from governance records alone — (1) configuration completeness, (2) conflict routing reproducibility, (3) resolution reproducibility, (4) evolution feed reproducibility, and (5) amendment reproducibility?

PASS indicates: All five requirements are satisfied. Every governance decision made during the FAI event traces to authored governance content sufficient to reproduce that decision. A reviewer examining only the governance records — without access to any party’s institutional memory, verbal understandings, or external documentation — can follow how each governance outcome was produced. The governance substrate is epistemically self-contained with respect to the decisions it governs.

FAIL indicates: Any one of the five requirements fails. The specific failing requirement identifies the governance quality gap precisely. Failure on requirement (1) means configuration records are incomplete — aspects of the FAI event were governed by configuration dimensions not captured in any record. Failure on requirement (2) means conflict routing decisions cannot be reproduced from the routing rules alone — routing depended on factors outside the authored rule set. Failure on requirement (3) means a resolved conflict cannot be traced from resolution outcome back through the resolution logic to the human authority whose decision produced it. Failure on requirement (4) means evolution feed routing cannot be verified from feed configuration records — what reached each participating Self’s evolution mechanisms is not reproducible from records. Failure on requirement (5) means amendment history cannot establish which version of any governance document was operative at which point during the event. Any of these failures triggers AP-22 (Black

Box Shared Substrate): governance decisions are not reproducible from records alone, and the shared substrate is opaque to the extent of the failure.

The determinism contract, as formalized in the derivation note series (D2.66), is not an independent governance requirement added on top of other quality requirements — it is a collected name for what it means for governance records to be substantively adequate. If all five determinism requirements are satisfied, the other governance quality commitments follow largely as corollaries: complete provenance fields (required by requirements 1 and 3), mutual accessibility of records (required for reproducibility to be achievable by any reviewer), consistent format (required for amendment reproducibility), and human authorization traceability (required for requirements 2, 3, and 5 to hold). This means a Test 41 pass gives strong evidence that quality is systemically present, while a Test 41 fail with a specific identified failing requirement gives a more diagnostic finding than “governance records are poor” — it identifies which governance quality property is missing and at which point in the governance chain.

3. Test 42 — Documentation Standards Compliance

Records to examine: All FAI governance records produced for the event.

Binary question: Do all governance records satisfy all four documentation standards — (1) non-specialist format accessibility, (2) complete provenance fields, (3) mutual accessibility by all participating governance practitioners, and (4) specified retention period?

PASS indicates: All four standards are satisfied for every record produced for the event. Non-specialist format accessibility means governance records are expressed in operational governance vocabulary, not in technical AI vocabulary that requires specialist expertise to parse. Complete provenance fields means every record carries the full set of required fields: author identity, timestamp, authority basis, and rationale. Mutual accessibility means no participating governance practitioner is structurally prevented from reading any record relevant to a governance decision that binds them. Specified retention period means every record carries an explicit designation of how long it is to be retained, consistent with the persistence policy the shared substrate’s configuration establishes.

FAIL indicates: Any standard fails for any record. Failure is specific — a finding names both the standard that failed and the record or records for which it failed. Failure on standard (1) means records exist in form but are expressed in vocabulary that excludes the practitioners they are supposed to govern. Failure on standard (2) means traceability is incomplete — some decisions cannot be followed back to an identified author with identified authority. Failure on standard (3) means governance is functionally unilateral despite being formally multi-party — some practitioners were bound by decisions they had no access to examine. Failure on standard (4) means audit continuity is uncertain — the evidentiary basis for any future review of the event is undefined. Any failure triggers AP-21 (Governance Theater): records exist in form but fail governance quality substance at the failing standard. The anti-pattern is named Governance Theater precisely because records that satisfy structural requirements but fail quality requirements produce the appearance of

governed operation without the substance.

4. Test 43 — Four-Link Authorization Chain Completeness

Records to examine: Contribution records (Link 1); FAI configuration authorization records (Link 2); shared-substrate governance records (Link 3); home absorption authorization records (Link 4).

Binary question: For any FAI-origin governance outcome — contributed content used in shared-substrate governance decisions, or content absorbed into a participating Self's home substrate — can all four authorization chain links be traced from the outcome back to the originating governance decision?

PASS indicates: The four-link chain is complete and navigable for all FAI-origin outcomes reviewed. Link 1 (contribution authorization) establishes that each participating Self's governance authorized the contribution of each aspect to the shared substrate. Link 2 (FAI configuration authorization) establishes that the configuration under which the event operated — sharing scope, cardinality, persistence policy, provenance carry-over depth — was itself authorized as substrate content under human authority before the event proceeded. Link 3 (shared-substrate governance) establishes that decisions made within the shared substrate during the event — conflict routing, resolution, escalation — operated under and are traceable to the orchestration rules and governance decisions that authorized them. Link 4 (home absorption authorization) establishes that each participating Self's governance authorized the specific content absorbed into its home substrate from the FAI event's evolution outputs. A complete chain means a reviewer can start at any FAI-origin outcome and navigate backward through all four links to the governance decisions that authorized each step, without any gap requiring inference, assumption, or external documentation.

FAIL indicates: Any link is absent for any outcome reviewed. A missing Link 1 means content entered the shared substrate without the contributing Self's governance having authorized the contribution. A missing Link 2 means the FAI event operated under configuration that was not itself authorized as substrate content — the meta-governance commitment (configuration as substrate content) was not satisfied. A missing Link 3 means shared-substrate decisions were made under rules or by processes not traceable to human-authored governance within the shared substrate. A missing Link 4 means content was absorbed into a home substrate without the receiving Self's governance having authorized that absorption. Any absence triggers AP-23 (Authorization Chain Gap): governance sovereignty cannot be proven for the affected outcome. When a link is missing, there is no record basis for establishing that the affected outcome is governed at all, regardless of whether the parties involved believe the outcome to be legitimate.

5. Test 44 — Non-Specialist Accessibility Verification

Records to examine: FAI governance records; governance role of the practitioners who authorized governance decisions.

Binary question: Could a governance practitioner without specialist AI technical expertise — using the governance records alone — make each governance authority decision (configuration review, escalation response, absorption authorization) without specialist assistance?

PASS indicates: All authority decisions are expressed in governance terms accessible to operational practitioners. Configuration review decisions can be made by a practitioner who understands what is being configured — which aspects are in scope, what persistence policy applies, what provenance depth is specified — without needing to understand how the AI substrate mediator implements those configurations. Escalation response decisions can be made by a practitioner who understands the governance stakes of the escalated conflict — what positions are in tension, what authority each position carries, what the governance consequences of each resolution path are — without needing AI technical expertise to evaluate the conflict. Absorption authorization decisions can be made by a practitioner who understands what content is being proposed for home substrate adoption and on what authority basis — without needing to evaluate the technical properties of the content as AI architecture.

FAIL indicates: Any authority decision requires specialist AI technical knowledge to evaluate. The failure identifies both the decision type and the specific technical knowledge requirement that makes it inaccessible. The non-specialist governance commitment (D2.75) is architectural, not organizational — it is a property of how governance records are constructed, not a property of which practitioners happen to have been hired. A governance record that expresses an authority decision in technical terms that require specialist interpretation has violated the accessibility commitment at the record level, regardless of whether a specialist happened to be available to assist. Governance structures can be formally committed to non-specialist accessibility while governance records are in practice expressed in vocabulary — model configuration parameters, substrate-mediator implementation details, technical conflict taxonomy — that only specialists can evaluate. When this occurs, governance authority decisions are in practice gated by specialist access. Test 44 catches this failure mode by asking whether records alone — not records plus specialist consultation — are sufficient for each authority decision.

Non-specialist governance (D2.75) commits that the three governance rights — inspect, modify, and override — are exercisable by non-specialist practitioners as a property of the system’s design. Test 44 is the operational verification that this design-time commitment is reflected in the quality of governance records produced at runtime. The commitment can be satisfied at design time and then systematically undermined by governance records that encode authority decisions in technical terms. Test 44 detects that undermining at the record level.

6. Test 45 — Non-Delegation Compliance

Records to examine: All governance decision records for the event; pre-authorization records where automated execution was used.

Binary question: For each governance authority decision — not merely labor — does a human governance authorization record exist, either as a direct authorization or as a pre-authorization that the automated execution references?

PASS indicates: All governance authority decisions have human authorization records. Where automated execution was used — an FAI event proceeding under configured orchestration rules without real-time human participation in each step — each automated action references a pre-authorization that a human governance authority produced before the event. The pre-authorization establishes that a human authored the rules under which the automation operated, approved the conditions under which those rules would govern the event, and established the scope within which the automation is authorized to act. The automated execution does not stand as self-authorizing; it stands as execution-under-authorization. The distinction between governance authority and governance labor is preserved throughout: humans may allocate labor to automated processes, but the authority that governs those processes traces to human-authored substrate content and human governance decisions recorded as such.

FAIL indicates: Any governance authority decision lacks a human authorization record. The failure identifies the specific decision and the absence — either no authorization record exists at all, or an automated action references no pre-authorization and therefore stands as self-authorizing. This triggers AP-24 (Automated Governance): governance authority has been impermissibly delegated. AP-24 is distinct from legitimate automation, which involves delegating labor under human-authored authority — it names cases where authority itself has been delegated, such that no human governance record authorizes or pre-authorizes the governance outcome. A FAI event that proceeds largely automatically under well-authored orchestration rules is not an AP-24 violation; a FAI event in which governance outcomes — configuration modifications, resolution decisions, absorption approvals — were produced by automated processes that reference no human authorization is.

7. Relationships among the five tests

The five tests are logically independent — a governance record can fail any subset of them while passing the others — but they are not unordered. Test 41 (determinism) has the broadest diagnostic reach: because the determinism contract collects the governance quality commitments that make records substantively adequate, a Test 41 pass provides the strongest evidence of overall quality, and a Test 41 fail with a specific identified failing sub-requirement provides a precise entry point for remediation.

Tests 42, 43, 44, and 45 then verify specific quality dimensions that the determinism contract requires but does not fully specify on its own. Test 42 verifies that records meet format and provenance standards that are preconditions for determinism to be assessable at all. Test 43 verifies that the authorization chain is navigable — a precondition for requirement (3) of the determinism contract to hold. Test 44 verifies that the accessibility commitment the architecture makes is operationally present in records — a quality property the determinism contract does not directly address. Test 45 verifies that human authorization is the actual source of governance authority — a property the determinism contract presupposes when it

requires that governance decisions trace to authored governance content.

In practice, a Test 41 fail often signals that one or more of Tests 42 through 45 will also fail, since the specific sub-requirement identified in Test 41 typically corresponds to a specific quality dimension one of the other tests addresses. Running all five tests provides full coverage and produces findings at the level of specificity needed to identify and remediate governance quality gaps.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model*. April 2026. ORCID: 0009-0004-8065-3235.

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